Appendix

Usage-based AI exposure across Australia's earnings distribution

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Every table here is read from pipeline output. File paths refer to the project repository. Differences in exposure are in exposure points (exposure × 100); rank gaps are in percentile points. Items added after the first results, including those added after review of the first draft, are labelled.

1 Alternative specifications

The main text uses one specification: weighting scheme 1, all units, EEH weekly earnings. This section reports every alternative, including those that do not hold under the rule in the Methods section.

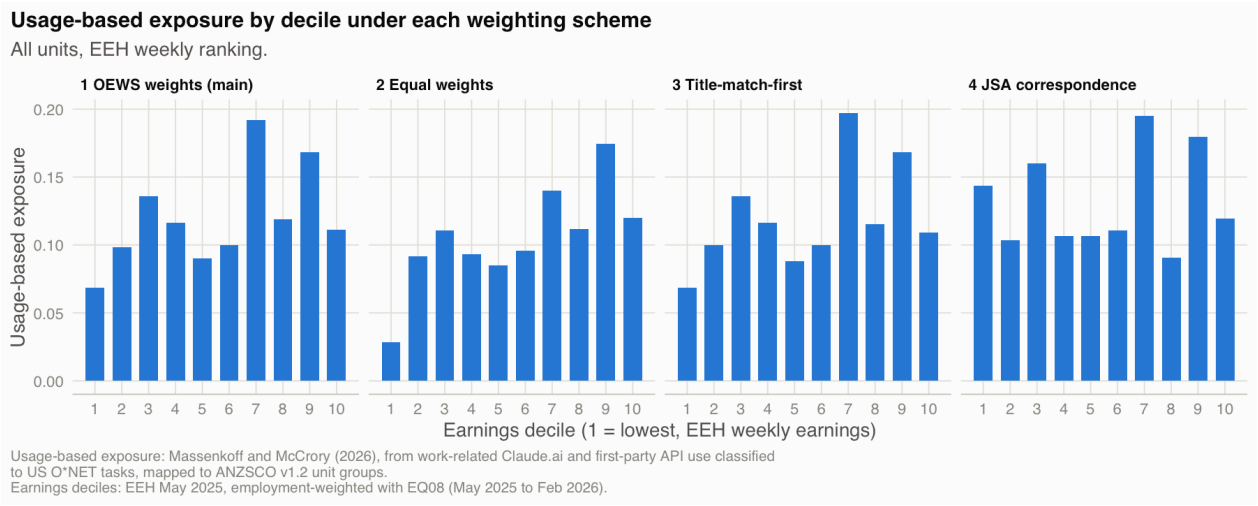


Figure 1: Usage-based exposure by decile under each weighting scheme.

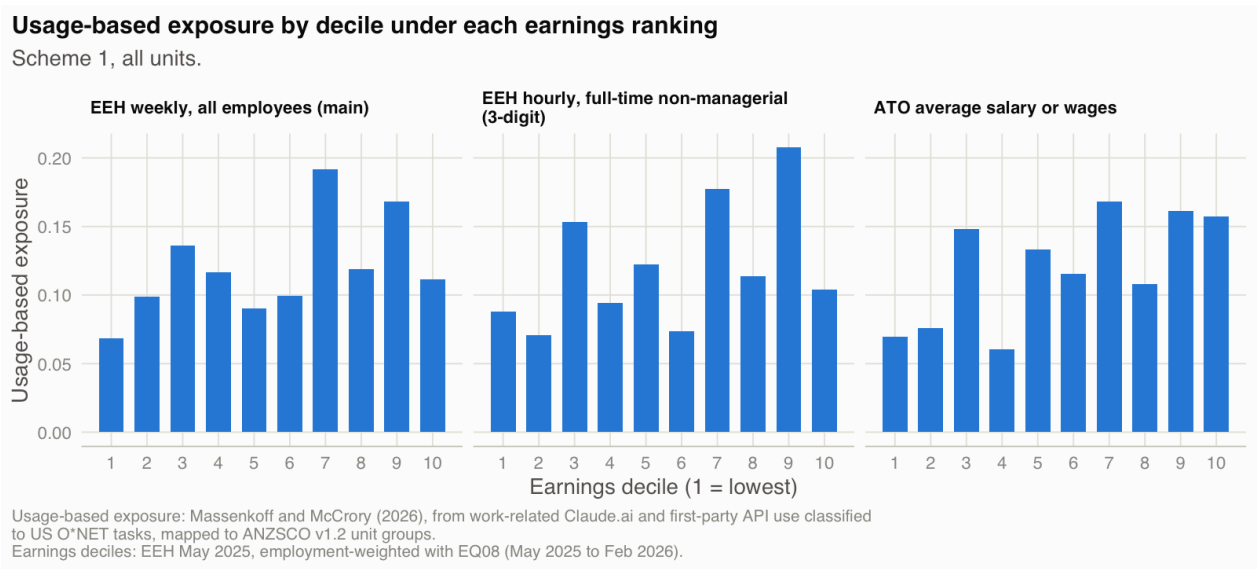


Figure 2: Usage-based exposure by decile under each earnings ranking.

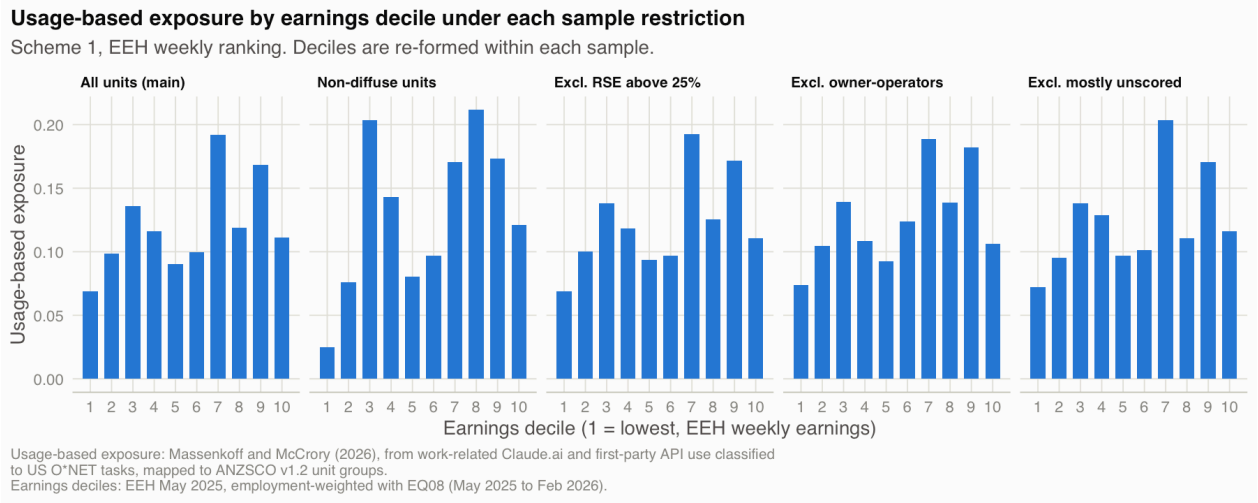


Figure 3: Usage-based exposure by decile under each sample restriction.

2 Sensitivity table

Decile profiles of usage-based exposure under every weighting scheme, sample and earnings ranking. "Holds" applies the rule in the Methods section against the primary specification (scheme 1, all units, EEH weekly earnings): the OLS slope of the decile means (shown, in exposure points per decile) has the main profile's sign, and the Spearman correlation with the main profile is at least 0.7. The concentration index (CI) and the rank index (rank CI, added after the first results) summarise the whole distribution.

Scheme	Sample	Ranking	Decile 1	Decile 10	Top minus bottom	Slope	Spearman with primary	CI	Rank CI	Holds
1 OEWS	All units	EEH weekly	0.069	0.111	4.3	0.62	1.00	0.089	0.073	yes
1 OEWS	All units	EEH hourly (cube 13)	0.088	0.104	1.6	0.67	0.73	0.093	0.085	yes
1 OEWS	All units	ATO salary	0.070	0.157	8.8	0.90	0.62	0.132	0.095	no
1 OEWS	Non-diffuse	EEH weekly	0.025	0.121	9.6	1.02	0.87	0.137	0.112	yes
1 OEWS	Non-diffuse	EEH hourly (cube 13)	0.050	0.111	6.1	0.91	0.89	0.114	0.104	yes
1 OEWS	Non-diffuse	ATO salary	0.034	0.162	12.8	1.49	0.50	0.192	0.143	no
1 OEWS	Excl. RSE above 25%	EEH weekly	0.069	0.111	4.2	0.63	0.99	0.089	0.075	yes
1 OEWS	Excl. RSE above 25%	EEH hourly (cube 13)	0.088	0.108	2.0	0.71	0.73	0.098	0.090	yes
1 OEWS	Excl. RSE above 25%	ATO salary	0.071	0.155	8.4	0.92	0.68	0.130	0.095	no
1 OEWS	Excl. owner-operators (draft)	EEH weekly	0.074	0.106	3.2	0.67	0.96	0.094	0.078	yes

Scheme	Sample	Ranking	Decile 1	Decile 10	Top minus bottom	Slope	Spearman with primary	CI	Rank CI	Holds
1 OEWS	Excl. owner-operators (draft)	EEH hourly (cube 13)	0.089	0.113	2.3	0.79	0.64	0.102	0.093	no
1 OEWS	Excl. owner-operators (draft)	ATO salary	0.074	0.144	7.0	0.95	0.82	0.130	0.096	yes
1 OEWS	Excl. mostly unscored (added after results)	EEH weekly	0.072	0.116	4.4	0.62	0.95	0.083	0.073	yes
1 OEWS	Excl. mostly unscored (added after results)	EEH hourly (cube 13)	0.087	0.107	2.0	0.72	0.65	0.093	0.086	no
1 OEWS	Excl. mostly unscored (added after results)	ATO salary	0.072	0.164	9.2	0.95	0.73	0.128	0.097	yes
2 Equal	All units	EEH weekly	0.028	0.120	9.2	0.94	0.88	0.152	0.096	yes
2 Equal	All units	EEH hourly (cube 13)	0.055	0.107	5.2	1.02	0.83	0.162	0.111	yes
2 Equal	All units	ATO salary	0.025	0.177	15.2	1.16	0.60	0.188	0.116	no
2 Equal	Non-diffuse	EEH weekly	0.021	0.140	11.9	1.14	0.79	0.165	0.121	yes
2 Equal	Non-diffuse	EEH hourly (cube 13)	0.030	0.129	9.9	1.11	0.93	0.156	0.122	yes
2 Equal	Non-diffuse	ATO salary	0.023	0.170	14.7	1.47	0.45	0.213	0.150	no
2 Equal	Excl. RSE above 25%	EEH weekly	0.028	0.120	9.2	0.96	0.89	0.153	0.099	yes
2 Equal	Excl. RSE above 25%	EEH hourly (cube 13)	0.056	0.111	5.5	1.06	0.83	0.168	0.117	yes
2 Equal	Excl. RSE above 25%	ATO salary	0.025	0.173	14.8	1.19	0.58	0.186	0.116	no
2 Equal	Excl. owner-operators (draft)	EEH weekly	0.030	0.114	8.4	1.02	0.82	0.160	0.107	yes

Scheme	Sample	Ranking	Decile 1	Decile 10	Top minus bottom	Slope	Spearman with primary	CI	Rank CI	Holds
2 Equal	Excl. owner-operators (draft)	EEH hourly (cube 13)	0.057	0.117	5.9	1.16	0.67	0.175	0.127	no
2 Equal	Excl. owner-operators (draft)	ATO salary	0.026	0.156	13.0	1.23	0.83	0.189	0.122	yes
2 Equal	Excl. mostly unscored (added after results)	EEH weekly	0.029	0.125	9.6	0.96	0.89	0.148	0.097	yes
2 Equal	Excl. mostly unscored (added after results)	EEH hourly (cube 13)	0.056	0.109	5.2	1.06	0.62	0.162	0.111	no
2 Equal	Excl. mostly unscored (added after results)	ATO salary	0.025	0.184	15.9	1.22	0.64	0.186	0.118	no
3 Title	All units	EEH weekly	0.069	0.109	4.1	0.60	0.98	0.087	0.065	yes
3 Title	All units	EEH hourly (cube 13)	0.088	0.102	1.4	0.63	0.73	0.089	0.074	yes
3 Title	All units	ATO salary	0.070	0.153	8.4	0.90	0.62	0.131	0.087	no
3 Title	Non-diffuse	EEH weekly	0.025	0.123	9.9	1.03	0.87	0.137	0.099	yes
3 Title	Non-diffuse	EEH hourly (cube 13)	0.050	0.113	6.4	0.90	0.83	0.113	0.090	yes
3 Title	Non-diffuse	ATO salary	0.033	0.164	13.0	1.52	0.50	0.196	0.130	no
3 Title	Excl. RSE above 25%	EEH weekly	0.069	0.109	4.0	0.61	0.99	0.087	0.067	yes
3 Title	Excl. RSE above 25%	EEH hourly (cube 13)	0.088	0.106	1.8	0.67	0.72	0.094	0.079	yes
3 Title	Excl. RSE above 25%	ATO salary	0.071	0.150	7.9	0.92	0.71	0.129	0.086	yes
3 Title	Excl. owner-operators (draft)	EEH weekly	0.074	0.104	3.0	0.65	0.94	0.091	0.071	yes

Scheme	Sample	Ranking	Decile 1	Decile 10	Top minus bottom	Slope	Spearman with primary	CI	Rank CI	Holds
3 Title	Excl. owner-operators (draft)	EEH hourly (cube 13)	0.089	0.110	2.1	0.76	0.64	0.097	0.084	no
3 Title	Excl. owner-operators (draft)	ATO salary	0.074	0.139	6.5	0.94	0.84	0.129	0.089	yes
3 Title	Excl. mostly unscored (added after results)	EEH weekly	0.072	0.114	4.2	0.60	0.93	0.080	0.065	yes
3 Title	Excl. mostly unscored (added after results)	EEH hourly (cube 13)	0.087	0.105	1.8	0.68	0.65	0.088	0.075	no
3 Title	Excl. mostly unscored (added after results)	ATO salary	0.072	0.159	8.7	0.94	0.76	0.128	0.089	yes
4 JSA links	All units	EEH weekly	0.144	0.119	−2.4	0.14	0.47	0.029	0.043	no
4 JSA links	All units	EEH hourly (cube 13)	0.138	0.111	−2.7	0.26	0.56	0.039	0.059	no
4 JSA links	All units	ATO salary	0.136	0.163	2.7	0.35	0.32	0.062	0.061	no
4 JSA links	Non-diffuse	EEH weekly	0.025	0.145	12.1	0.94	0.84	0.129	0.098	yes
4 JSA links	Non-diffuse	EEH hourly (cube 13)	0.033	0.136	10.3	0.88	0.77	0.119	0.099	yes
4 JSA links	Non-diffuse	ATO salary	0.022	0.155	13.3	1.34	0.49	0.182	0.127	no
4 JSA links	Excl. RSE above 25%	EEH weekly	0.145	0.120	−2.5	0.14	0.47	0.028	0.044	no
4 JSA links	Excl. RSE above 25%	EEH hourly (cube 13)	0.136	0.115	−2.1	0.30	0.54	0.044	0.062	no
4 JSA links	Excl. RSE above 25%	ATO salary	0.139	0.159	2.0	0.38	0.36	0.059	0.060	no
4 JSA links	Excl. owner-operators (draft)	EEH weekly	0.155	0.117	−3.9	0.16	0.60	0.033	0.048	no

Scheme	Sample	Ranking	Decile 1	Decile 10	Top minus bottom	Slope	Spearman with primary	CI	Rank CI	Holds
4 JSA links	Excl. owner-operators (draft)	EEH hourly (cube 13)	0.140	0.122	−1.8	0.43	0.38	0.052	0.070	no
4 JSA links	Excl. owner-operators (draft)	ATO salary	0.146	0.142	−0.4	0.39	0.49	0.063	0.064	no
4 JSA links	Excl. mostly unscored (added after results)	EEH weekly	0.152	0.127	−2.5	0.26	0.39	0.037	0.048	no
4 JSA links	Excl. mostly unscored (added after results)	EEH hourly (cube 13)	0.130	0.117	−1.3	0.37	0.35	0.045	0.061	no
4 JSA links	Excl. mostly unscored (added after results)	ATO salary	0.143	0.172	2.8	0.54	0.52	0.073	0.067	no

3 The profile without decile boundaries

Added after review of the first draft. Decile means can exaggerate single-unit spikes, so the profile is also shown as a smoothed curve over earnings percentiles.

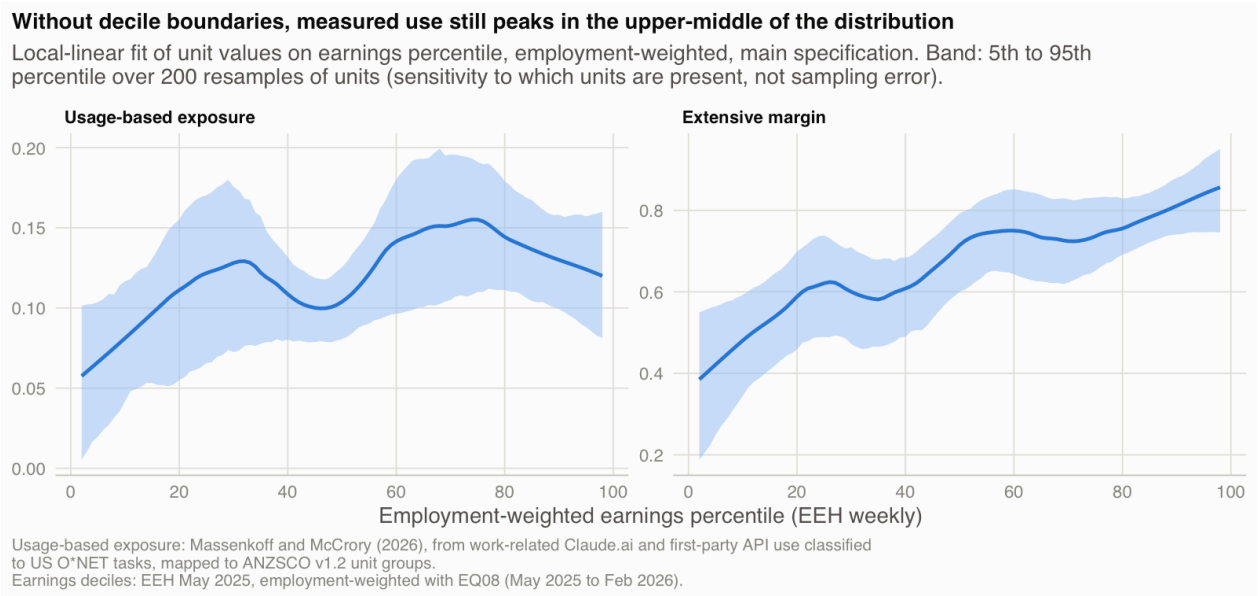


Figure 4: Smoothed profile over earnings percentiles, main specification.

4 Margins, lower bound and capability analogue by decile

Main specification. The log decomposition compares each decile with decile 1: $\ln(e_d/e_1) = \ln(m_d/m_1) + \ln(i_d/i_1)$. The lower bound counts linked US occupations without a score as zero instead of renormalising over scored ones. The capability analogue applies the same weights to an indicator of a positive Eloundou et al. beta. All three were added after review of the first draft.

Decile	Exposure	Lower bound	Extensive margin	Intensity	Share of linked occupations with beta > 0	Extensive share of log difference from decile 1
1	0.069	0.064	0.388	0.177	0.866	
2	0.099	0.094	0.561	0.176	0.864	102%
3	0.136	0.124	0.672	0.202	0.975	80%
4	0.116	0.112	0.555	0.210	0.978	68%
5	0.090	0.082	0.658	0.137	0.970	192%
6	0.100	0.092	0.793	0.126	0.994	192%
7	0.192	0.182	0.695	0.276	0.987	57%
8	0.119	0.105	0.668	0.178	0.983	99%
9	0.168	0.140	0.860	0.196	0.998	89%
10	0.111	0.103	0.814	0.137	0.972	153%

5 Concentration indices for every measure

Main specification. The level index $2\,\mathrm{cov}(e, r)/\bar{e}$ depends on each measure's mean and is not comparable across measures. The rank index $4\,\mathrm{cov}(R_e, r)$ is; without ties it equals one third of the weighted Spearman correlation with earnings. The Erreygers index applies to measures bounded on $[0, 1]$. The generalised index $2\,\mathrm{cov}(e, r)$ is in the measure's own units. AIOE takes negative values, so its level and Erreygers indices are not defined. Rank, Erreygers and generalised indices were added after review of the first draft.

Measure	Level	Rank	Erreygers	Generalised	Employment covered
Usage-based exposure	0.089	0.073	0.043	0.0107	99.1%
Usage-based exposure, lower bound	0.078	0.076	0.034	0.0085	99.1%
Extensive margin	0.097	0.084	0.259	0.0649	99.1%
Share of linked occupations with beta > 0	0.022	0.063	0.083	0.0208	100.0%
JSA automation	0.013	0.035	0.020	0.0049	100.0%
JSA augmentation	0.037	0.147	0.094	0.0236	100.0%
Eloundou beta	0.116	0.137	0.167	0.0418	100.0%
AIOE		0.141		0.2198	100.0%

6 Usage-potential gap: bottom and top quintiles

Mean rank gap (percentile points) in the bottom and top earnings quintiles, all units. AIOE appears only in this aggregate form.

Potential	Ranking	Scheme	Bottom quintile	Top quintile
JSA automation	EEH weekly	1 OEWS	−5.1	3.8
JSA automation	EEH weekly	2 Equal	−9.1	7.4
JSA automation	EEH weekly	3 Title	−4.1	2.9
JSA automation	EEH weekly	4 JSA links	1.6	4.0
JSA automation	EEH hourly (cube 13)	1 OEWS	−3.5	−0.4

Potential	Ranking	Scheme	Bottom quintile	Top quintile
JSA automation	EEH hourly (cube 13)	2 Equal	−8.1	4.6
JSA automation	EEH hourly (cube 13)	3 Title	−2.3	−1.4
JSA automation	EEH hourly (cube 13)	4 JSA links	0.3	−0.9
JSA automation	ATO salary	1 OEWS	−5.3	5.3
JSA automation	ATO salary	2 Equal	−9.6	7.6
JSA automation	ATO salary	3 Title	−4.3	4.8
JSA automation	ATO salary	4 JSA links	1.3	3.7
JSA augmentation	EEH weekly	1 OEWS	7.7	−10.8
JSA augmentation	EEH weekly	2 Equal	3.7	−7.2
JSA augmentation	EEH weekly	3 Title	8.8	−11.8
JSA augmentation	EEH weekly	4 JSA links	14.7	−11.0
JSA augmentation	EEH hourly (cube 13)	1 OEWS	9.9	−13.3
JSA augmentation	EEH hourly (cube 13)	2 Equal	5.4	−8.3
JSA augmentation	EEH hourly (cube 13)	3 Title	11.0	−14.5
JSA augmentation	EEH hourly (cube 13)	4 JSA links	14.4	−14.2
JSA augmentation	ATO salary	1 OEWS	8.4	−8.4
JSA augmentation	ATO salary	2 Equal	4.1	−6.0
JSA augmentation	ATO salary	3 Title	9.5	−9.0
JSA augmentation	ATO salary	4 JSA links	15.2	−10.1
Eloundou beta	EEH weekly	1 OEWS	8.4	−8.5
Eloundou beta	EEH weekly	2 Equal	2.8	−4.8
Eloundou beta	EEH weekly	3 Title	8.8	−9.5
Eloundou beta	EEH weekly	4 JSA links	10.9	−7.7
Eloundou beta	EEH hourly (cube 13)	1 OEWS	10.4	−8.7
Eloundou beta	EEH hourly (cube 13)	2 Equal	4.2	−5.0
Eloundou beta	EEH hourly (cube 13)	3 Title	10.9	−9.8
Eloundou beta	EEH hourly (cube 13)	4 JSA links	10.9	−10.7
Eloundou beta	ATO salary	1 OEWS	8.5	−8.5
Eloundou beta	ATO salary	2 Equal	2.9	−6.4
Eloundou beta	ATO salary	3 Title	9.0	−9.0
Eloundou beta	ATO salary	4 JSA links	11.0	−9.7
AIOE	EEH weekly	1 OEWS	5.7	−8.5
AIOE	EEH weekly	2 Equal	1.7	−4.8
AIOE	EEH weekly	3 Title	6.7	−9.5
AIOE	EEH weekly	4 JSA links	12.1	−8.7
AIOE	EEH hourly (cube 13)	1 OEWS	10.5	−11.9
AIOE	EEH hourly (cube 13)	2 Equal	5.9	−6.9
AIOE	EEH hourly (cube 13)	3 Title	11.5	−13.1
AIOE	EEH hourly (cube 13)	4 JSA links	14.3	−12.8

Potential	Ranking	Scheme	Bottom quintile	Top quintile
AIOE	ATO salary	1 OEWS	5.7	−11.0
AIOE	ATO salary	2 Equal	1.4	−8.6
AIOE	ATO salary	3 Title	6.7	−11.6
AIOE	ATO salary	4 JSA links	12.0	−12.9

Potential	Ranking	Bottom sign same in all schemes	Top sign same in all schemes
AIOE	ATO salary	yes	yes
Eloundou beta	ATO salary	yes	yes
JSA augmentation	ATO salary	yes	yes
JSA automation	ATO salary	no	yes
AIOE	EEH weekly	yes	yes
Eloundou beta	EEH weekly	yes	yes
JSA augmentation	EEH weekly	yes	yes
JSA automation	EEH weekly	no	yes
AIOE	EEH hourly (cube 13)	yes	yes
Eloundou beta	EEH hourly (cube 13)	yes	yes
JSA augmentation	EEH hourly (cube 13)	yes	yes
JSA automation	EEH hourly (cube 13)	no	no

Without zero-exposure units. Added after review of the first draft. All units with zero usage-based exposure share one tied rank, so their gaps are set by the potential measure alone. Here the gaps are recomputed among units with positive exposure, within the same deciles (EEH weekly ranking). Averaging over random tie-breaks would give exactly the mid-rank results above, because the expected rank under random tie-breaking is the mid-rank.

Potential	Scheme	Bottom quintile	Top quintile
JSA automation	1 OEWS	−5.1	3.6
JSA automation	2 Equal	−12.1	8.1
JSA automation	3 Title	−4.5	4.1
JSA automation	4 JSA links	2.9	2.5
JSA augmentation	1 OEWS	10.6	−13.2
JSA augmentation	2 Equal	3.6	−8.8
JSA augmentation	3 Title	11.2	−14.0
JSA augmentation	4 JSA links	24.0	−18.0
Eloundou beta	1 OEWS	12.1	−10.3
Eloundou beta	2 Equal	2.8	−6.1
Eloundou beta	3 Title	11.8	−10.2
Eloundou beta	4 JSA links	17.9	−11.6
AIOE	1 OEWS	9.8	−10.4
AIOE	2 Equal	2.9	−5.9
AIOE	3 Title	10.1	−10.0
AIOE	4 JSA links	24.9	−12.1

7 Influence of single large units

The top-minus-bottom difference, decile 7 (the peak), and the concentration indices with each of the five largest units in deciles 1, 7 and 10 left out in turn. Deciles are re-formed each time. “Holds” applies the rule against the main profile. This check was added after the first results and extended to decile 7, the indices and the rule after review of the first draft.

Scheme	Left out	From decile	Top minus bottom	Decile 7	CI	Rank CI	Holds
1 OEWS	(none)		4.3	0.192	0.089	0.073	yes
1 OEWS	6211 Sales Assistants (General)	1	6.9	0.216	0.098	0.090	yes
1 OEWS	8513 Kitchenhands	1	4.1	0.191	0.080	0.067	yes
1 OEWS	4315 Waiters	1	4.1	0.191	0.081	0.068	yes
1 OEWS	6311 Checkout Operators and Office Cashiers	1	4.8	0.191	0.088	0.076	yes
1 OEWS	8912 Shelf Fillers	1	4.2	0.195	0.084	0.070	yes
1 OEWS	2211 Accountants	7	4.1	0.172	0.085	0.072	yes
1 OEWS	7331 Truck Drivers	7	4.2	0.219	0.093	0.078	yes
1 OEWS	2251 Advertising and Marketing Professionals	7	4.2	0.161	0.080	0.071	yes
1 OEWS	2231 Human Resource Professionals	7	4.2	0.190	0.088	0.073	yes
1 OEWS	3223 Structural Steel and Welding Trades Workers	7	4.2	0.202	0.090	0.074	yes
1 OEWS	1311 Advertising, Public Relations and Sales Managers	10	3.1	0.166	0.088	0.070	yes
1 OEWS	1331 Construction Managers	10	4.3	0.174	0.094	0.074	yes
1 OEWS	1323 Human Resource Managers	10	4.3	0.181	0.093	0.074	yes
1 OEWS	1351 ICT Managers	10	3.7	0.182	0.088	0.072	yes
1 OEWS	1322 Finance Managers	10	2.5	0.185	0.079	0.070	yes
2 Equal	(none)		9.2	0.140	0.152	0.096	yes
2 Equal	6211 Sales Assistants (General)	1	8.2	0.165	0.133	0.091	yes
2 Equal	8513 Kitchenhands	1	9.0	0.142	0.144	0.090	yes
2 Equal	4315 Waiters	1	9.0	0.142	0.145	0.091	yes
2 Equal	6311 Checkout Operators and Office Cashiers	1	9.4	0.142	0.149	0.096	yes
2 Equal	8912 Shelf Fillers	1	9.1	0.144	0.148	0.093	yes
2 Equal	2211 Accountants	7	9.1	0.144	0.152	0.096	yes
2 Equal	7331 Truck Drivers	7	9.1	0.168	0.156	0.102	yes
2 Equal	2251 Advertising and Marketing Professionals	7	9.2	0.122	0.147	0.094	yes
2 Equal	2231 Human Resource Professionals	7	9.2	0.141	0.152	0.096	yes
2 Equal	3223 Structural Steel and Welding Trades Workers	7	9.2	0.150	0.153	0.097	yes
2 Equal	1311 Advertising, Public Relations and Sales Managers	10	7.6	0.124	0.148	0.092	yes
2 Equal	1331 Construction Managers	10	9.1	0.129	0.156	0.095	yes
2 Equal	1323 Human Resource Managers	10	8.8	0.133	0.153	0.095	yes

Scheme	Left out	From decile	Top minus bottom	Decile 7	CI	Rank CI	Holds
2 Equal	1351 ICT Managers	10	8.6	0.133	0.152	0.095	yes
2 Equal	1322 Finance Managers	10	7.5	0.135	0.142	0.093	yes
3 Title	(none)		4.1	0.197	0.087	0.065	yes
3 Title	6211 Sales Assistants (General)	1	6.7	0.223	0.095	0.081	yes
3 Title	8513 Kitchenhands	1	3.9	0.197	0.078	0.059	yes
3 Title	4315 Waiters	1	3.9	0.197	0.079	0.060	yes
3 Title	6311 Checkout Operators and Office Cashiers	1	4.6	0.197	0.085	0.067	yes
3 Title	8912 Shelf Fillers	1	4.0	0.200	0.082	0.062	yes
3 Title	2211 Accountants	7	3.9	0.177	0.082	0.063	yes
3 Title	7331 Truck Drivers	7	3.9	0.225	0.090	0.070	yes
3 Title	2251 Advertising and Marketing Professionals	7	4.0	0.165	0.078	0.063	yes
3 Title	2231 Human Resource Professionals	7	4.0	0.195	0.086	0.065	yes
3 Title	3223 Structural Steel and Welding Trades Workers	7	4.0	0.208	0.087	0.066	yes
3 Title	1311 Advertising, Public Relations and Sales Managers	10	2.9	0.170	0.086	0.062	yes
3 Title	1331 Construction Managers	10	3.5	0.179	0.088	0.065	yes
3 Title	1323 Human Resource Managers	10	4.5	0.186	0.093	0.071	yes
3 Title	1351 ICT Managers	10	3.5	0.187	0.086	0.063	yes
3 Title	1322 Finance Managers	10	2.3	0.190	0.077	0.062	yes
4 JSA links	(none)		−2.4	0.195	0.029	0.043	no
4 JSA links	6211 Sales Assistants (General)	1	7.7	0.223	0.092	0.075	yes
4 JSA links	8513 Kitchenhands	1	−2.6	0.198	0.019	0.037	no
4 JSA links	4315 Waiters	1	−2.6	0.198	0.022	0.039	no
4 JSA links	6311 Checkout Operators and Office Cashiers	1	−1.9	0.198	0.026	0.045	no
4 JSA links	8912 Shelf Fillers	1	−2.5	0.198	0.024	0.040	no
4 JSA links	2211 Accountants	7	−2.6	0.169	0.023	0.042	no
4 JSA links	7331 Truck Drivers	7	−2.5	0.218	0.033	0.047	no
4 JSA links	2251 Advertising and Marketing Professionals	7	−2.5	0.162	0.019	0.041	no
4 JSA links	2231 Human Resource Professionals	7	−2.5	0.183	0.026	0.042	no
4 JSA links	3223 Structural Steel and Welding Trades Workers	7	−2.5	0.204	0.030	0.044	no
4 JSA links	1311 Advertising, Public Relations and Sales Managers	10	−5.5	0.165	0.015	0.039	no
4 JSA links	1331 Construction Managers	10	−2.6	0.174	0.034	0.044	no
4 JSA links	1323 Human Resource Managers	10	−2.2	0.183	0.035	0.048	no
4 JSA links	1351 ICT Managers	10	−3.6	0.184	0.025	0.042	no
4 JSA links	1322 Finance Managers	10	−4.4	0.187	0.019	0.040	no

8 Rank correlations

Employment-weighted Spearman correlations between usage-based exposure and each potential measure, by weighting scheme. Under scheme 4, 354 units have links and 347 have a scored link.

Scheme	Potential	Units	Weighted	Unweighted
1 OEWS	JSA automation	354	0.72	0.68
1 OEWS	JSA augmentation	354	0.64	0.66
1 OEWS	Eloundou beta	355	0.77	0.79
1 OEWS	AIOE	353	0.79	0.78
2 Equal	JSA automation	354	0.72	0.72
2 Equal	JSA augmentation	354	0.69	0.70
2 Equal	Eloundou beta	355	0.84	0.83
2 Equal	AIOE	353	0.84	0.80
3 Title	JSA automation	351	0.71	0.67
3 Title	JSA augmentation	351	0.62	0.64
3 Title	Eloundou beta	352	0.75	0.76
3 Title	AIOE	350	0.76	0.74
4 JSA links	JSA automation	347	0.74	0.70
4 JSA links	JSA augmentation	347	0.64	0.66
4 JSA links	Eloundou beta	347	0.78	0.78
4 JSA links	AIOE	347	0.71	0.74

9 Largest rank disagreements with JSA automation

The fifteen units with the largest gap under scheme 1 whose sign is the same under all four schemes. The scheme columns give the gap, in percentile points, under each weighting scheme. A negative gap means usage-based exposure ranks the unit lower than JSA automation does. “Contribution” is the scheme-1 gap times the unit’s share of employment, in percentile points (added after the first results).

Unit	Title	Employment (‘000)	Usage-based exposure	JSA automation	Scheme 1	Scheme 2	Scheme 3	Scheme 4	Contribution
5611	Betting Clerks	1.5	0.007	0.71	−77.1	−70.3	−74.7	−81.7	−0.01
5912	Transport and Despatch Clerks	45.2	0.012	0.64	−69.1	−41.3	−66.5	−44.0	−0.21
5911	Purchasing and Supply Logistics Clerks	121.6	0.009	0.58	−66.4	−56.3	−63.6	−48.6	−0.55
1411	Cafe and Restaurant Managers	61.3	0.000†	0.43	−60.9	−60.9	−59.3	−51.2	−0.25
8991	Caretakers	4.5	0.000†	0.43	−60.9	−60.9	−59.3	−51.2	−0.02
5521	Bank Workers	33.5	0.019	0.54	−59.8	−60.8	−56.9	−43.2	−0.14

Unit	Title	Employment ('000)	Usage-based exposure	JSA automation	Scheme 1	Scheme 2	Scheme 3	Scheme 4	Contribution
1211	Aquaculture Farmers	2.2	0.000†	0.41	−54.0	−54.0	−52.4	−44.0	−0.01
7321	Delivery Drivers	82.5	0.007	0.47	−53.9	−57.6	−51.3	−57.0	−0.30
2113	Photographers	13.1	0.195	0.24	52.8	54.8	52.8	48.0	0.05
7312	Bus and Coach Drivers	40.0	0.000†	0.39	−51.3	−51.3	−49.6	−41.2	−0.14
3422	Electrical Distribution Trades Workers	11.9	0.084	0.15	51.2	46.5	50.7	10.8	0.04
1334	Manufacturers	19.2	0.013	0.47	−49.5	−48.1	−46.9	−34.1	−0.06
7313	Train and Tram Drivers	19.7	0.000†	0.38	−49.4	−49.4	−47.7	−39.2	−0.07
5513	Payroll Clerks	46.1	0.049	0.68	−49.0	−43.8	−48.2	−46.5	−0.15
2114	Visual Arts and Crafts Professionals	16.3	0.075	0.16	48.3	58.0	47.9	69.5	0.05

† Below the usage threshold: a score of exactly zero.

Ranked by contribution. The fifteen sign-stable disagreements with the largest employment-weighted contribution. Added after review of the first draft, because ranking by the gap alone puts small units first.

Unit	Title	Employment ('000)	Usage-based exposure	JSA automation	Scheme 1	Scheme 2	Scheme 3	Scheme 4	Contribution
5911	Purchasing and Supply Logistics Clerks	121.6	0.009	0.58	−66.4	−56.3	−63.6	−48.6	−0.55
2414	Secondary School Teachers	210.1	0.290	0.35	35.3	38.7	35.6	28.4	0.51
2412	Primary School Teachers	205.2	0.163	0.32	31.1	40.4	31.1	21.5	0.44
3411	Electricians	199.0	0.037	0.16	31.2	13.1	33.7	9.3	0.42
2544	Registered Nurses	367.4	0.062	0.30	15.7	18.2	15.6	20.1	0.39
7411	Storepersons	166.7	0.002	0.34	−33.1	−18.3	−30.2	−31.0	−0.38
7321	Delivery Drivers	82.5	0.007	0.47	−53.9	−57.6	−51.3	−57.0	−0.30
1411	Cafe and Restaurant Managers	61.3	0.000†	0.43	−60.9	−60.9	−59.3	−51.2	−0.25

Unit	Title	Employment ('000)	Usage-based exposure	JSA automation	Scheme 1	Scheme 2	Scheme 3	Scheme 4	Contribution
3121	Architectural, Building and Surveying Technicians	79.8	0.031	0.51	−46.3	−34.4	−43.9	−37.3	−0.25
3622	Gardeners	85.8	0.066	0.19	42.2	31.6	41.3	4.9	0.25
4117	Welfare Support Workers	90.5	0.014	0.42	−39.6	−38.0	−37.0	−22.7	−0.24
1421	Retail Managers	247.9	0.127	0.48	−13.7	−10.0	−13.3	−1.7	−0.23
4315	Waiters	137.1	0.000†	0.28	−24.1	−24.1	−22.7	−13.5	−0.23
1213	Livestock Farmers	80.1	0.000†	0.34	−41.1	−41.1	−39.4	−31.0	−0.22
5912	Transport and Despatch Clerks	45.2	0.012	0.64	−69.1	−41.3	−66.5	−44.0	−0.21

10 Women and men: within and between deciles

Added after review of the first draft. The difference between women's and men's mean usage-based exposure is split into a within-decile part, which weights the decile differences by the average of women's and men's employment shares, and a between-decile part, which weights the difference in shares by the average of the two means. The parts add to the total.

Women minus men	Within deciles	Between deciles	Within share
0.0425	0.0546	−0.0121	128%

11 Western Australia: contributions by major group

WA minus national mean usage-based exposure (× 100), by major group. "Raw" is $\sum_{a \in g}(s_a^{WA} - s_a^{AUS})e_a$, as in the first draft; "relative to the national mean" subtracts the national mean from e_a and does not depend on the level of the measure. Both sum to the same total. The normalised version was added after review of the first draft.

Major group	WA share	National share	Raw	Relative to the national mean
Managers	11.7%	12.8%	−0.20	−0.07
Professionals	25.2%	27.8%	−0.96	−0.64
Technicians and Trades Workers	16.7%	13.6%	0.05	−0.32
Community and Personal Service Workers	10.0%	11.1%	−0.02	0.11
Clerical and Administrative Workers	11.9%	12.6%	−0.23	−0.15
Sales Workers	7.5%	7.6%	−0.02	−0.01
Machinery Operators and Drivers	8.2%	6.2%	0.00	−0.24
Labourers	8.9%	8.4%	0.01	−0.05

Measure	WA	Australia	Difference	Relative difference
Usage-based exposure	0.107	0.120	−0.0137	−11.4%
Extensive margin	0.627	0.667	−0.0406	−6.1%

Measure	WA	Australia	Difference	Relative difference
JSA automation	0.362	0.373	−0.0118	−3.2%
Eloundou beta	0.344	0.360	−0.0165	−4.6%

12 Usage relative to employment: published and rescaled shares

The published SOC major-group shares do not sum to one hundred per cent (see the last tables of this appendix). “Rescaled” divides each geography and month by its published total before taking the ratio to employment. Added after review of the first draft.

Geography	SOC major group	Published	Rescaled
AU-WA	Computer and Mathematical	13.94	14.22
AU-WA	Arts, Design, Entertainment, Sports, and Media	10.37	10.57
AU-WA	Life, Physical, and Social Science	2.74	2.79
AU-WA	Educational Instruction and Library	2.28	2.32
AU-WA	Sales and Related	1.62	1.66
AU-WA	Community and Social Service	1.53	1.56
AU-WA	Legal	1.30	1.32
AU-WA	Architecture and Engineering	1.01	1.03
AU-WA	Office and Administrative Support	0.88	0.90
AU-WA	Business and Financial Operations	0.71	0.72
AU-WA	Management	0.68	0.70
AU-WA	Healthcare Practitioners and Technical	0.54	0.55
AU-WA	Personal Care and Service	0.41	0.42
AU-WA	Production	0.27	0.28
AU-WA	Protective Service	0.22	0.22
AU-WA	Healthcare Support	0.18	0.18
AU-WA	Installation, Maintenance, and Repair	0.14	0.14
AU-WA	Food Preparation and Serving Related	0.07	0.07
AU-WA	Transportation and Material Moving	0.04	0.04
AUS	Computer and Mathematical	8.55	8.70
AUS	Arts, Design, Entertainment, Sports, and Media	7.88	8.02
AUS	Life, Physical, and Social Science	3.55	3.61
AUS	Educational Instruction and Library	2.05	2.09
AUS	Sales and Related	1.63	1.66
AUS	Community and Social Service	1.56	1.59
AUS	Architecture and Engineering	1.28	1.31
AUS	Legal	0.93	0.95
AUS	Office and Administrative Support	0.84	0.85
AUS	Business and Financial Operations	0.62	0.63
AUS	Management	0.60	0.61
AUS	Healthcare Practitioners and Technical	0.51	0.52
AUS	Personal Care and Service	0.37	0.38

Geography	SOC major group	Published	Rescaled
AUS	Production	0.29	0.30
AUS	Protective Service	0.22	0.22
AUS	Installation, Maintenance, and Repair	0.18	0.19
AUS	Healthcare Support	0.15	0.15
AUS	Food Preparation and Serving Related	0.07	0.07
AUS	Building and Grounds Cleaning and Maintenance	0.05	0.05
AUS	Transportation and Material Moving	0.04	0.04
AUS	Farming, Fishing, and Forestry	0.02	0.02
AUS	Construction and Extraction	0.02	0.02
GLOBAL	Arts, Design, Entertainment, Sports, and Media	10.08	10.43
GLOBAL	Computer and Mathematical	7.06	7.31
GLOBAL	Life, Physical, and Social Science	4.78	4.94
GLOBAL	Architecture and Engineering	2.11	2.19
GLOBAL	Educational Instruction and Library	2.11	2.18
GLOBAL	Community and Social Service	1.46	1.51
GLOBAL	Legal	1.16	1.19
GLOBAL	Sales and Related	1.01	1.04
GLOBAL	Business and Financial Operations	0.83	0.86
GLOBAL	Management	0.81	0.83
GLOBAL	Office and Administrative Support	0.67	0.69
GLOBAL	Personal Care and Service	0.57	0.59
GLOBAL	Healthcare Practitioners and Technical	0.49	0.51
GLOBAL	Production	0.19	0.19
GLOBAL	Installation, Maintenance, and Repair	0.15	0.16
GLOBAL	Protective Service	0.14	0.14
GLOBAL	Farming, Fishing, and Forestry	0.12	0.13
GLOBAL	Healthcare Support	0.12	0.12
GLOBAL	Food Preparation and Serving Related	0.05	0.05
GLOBAL	Building and Grounds Cleaning and Maintenance	0.04	0.05
GLOBAL	Transportation and Material Moving	0.04	0.04
GLOBAL	Construction and Extraction	0.02	0.02

13 Another platform: Copilot-based applicability

Added for v1.0, after the first results. Microsoft's AI applicability score (Tomlinson et al. 2025, CC BY 4.0) is mapped to ANZSCO through the same links and weights as the usage-based score. Measures are compared on employment-weighted ranks. The composite is the mean of each unit's two percentile ranks; with two sources the median equals the mean.

Measure	Units	Rank index	Rank correlation with Claude usage	Rank correlation with earnings
Claude usage-based exposure	355	0.073	1.00	0.22
Copilot AI applicability	355	−0.011	0.65	−0.03

Measure	Units	Rank index	Rank correlation with Claude usage	Rank correlation with earnings
JSA automation	357	0.035	0.72	0.10
JSA augmentation	357	0.147	0.64	0.44
Eloundou beta	357	0.137	0.77	0.41
AIOE	355	0.141	0.79	0.42

Every specification: rank indices on the units with both measures.

Scheme	Sample	Ranking	Units	Claude	Copilot	Composite	Rank correlation
1 OEWS	All units	EEH weekly	355	0.073	−0.011	0.033	0.65
1 OEWS	All units	EEH hourly (cube 13)	352	0.085	0.025	0.058	0.66
1 OEWS	All units	ATO salary	355	0.095	0.009	0.056	0.65
1 OEWS	Non-diffuse	EEH weekly	214	0.112	−0.007	0.059	0.60
1 OEWS	Non-diffuse	EEH hourly (cube 13)	213	0.104	0.013	0.065	0.61
1 OEWS	Non-diffuse	ATO salary	214	0.143	0.026	0.095	0.60
1 OEWS	Excl. RSE above 25%	EEH weekly	330	0.075	−0.012	0.034	0.65
1 OEWS	Excl. RSE above 25%	EEH hourly (cube 13)	327	0.090	0.028	0.063	0.65
1 OEWS	Excl. RSE above 25%	ATO salary	330	0.095	0.007	0.055	0.65
1 OEWS	Excl. owner-operators (draft)	EEH weekly	342	0.078	−0.012	0.036	0.65
1 OEWS	Excl. owner-operators (draft)	EEH hourly (cube 13)	339	0.093	0.029	0.066	0.65
1 OEWS	Excl. owner-operators (draft)	ATO salary	342	0.096	0.001	0.053	0.65
1 OEWS	Excl. mostly unscored (added after results)	EEH weekly	332	0.073	0.002	0.043	0.69
1 OEWS	Excl. mostly unscored (added after results)	EEH hourly (cube 13)	329	0.086	0.035	0.066	0.69
1 OEWS	Excl. mostly unscored (added after results)	ATO salary	332	0.097	0.023	0.067	0.69
2 Equal	All units	EEH weekly	355	0.096	0.016	0.060	0.72
2 Equal	All units	EEH hourly (cube 13)	352	0.111	0.047	0.085	0.72
2 Equal	All units	ATO salary	355	0.116	0.037	0.082	0.72
2 Equal	Non-diffuse	EEH weekly	214	0.121	0.008	0.070	0.67
2 Equal	Non-diffuse	EEH hourly (cube 13)	213	0.122	0.023	0.079	0.67
2 Equal	Non-diffuse	ATO salary	214	0.150	0.046	0.107	0.67
2 Equal	Excl. RSE above 25%	EEH weekly	330	0.099	0.016	0.062	0.71
2 Equal	Excl. RSE above 25%	EEH hourly (cube 13)	327	0.117	0.051	0.091	0.71
2 Equal	Excl. RSE above 25%	ATO salary	330	0.116	0.036	0.082	0.71
2 Equal	Excl. owner-operators (draft)	EEH weekly	342	0.107	0.022	0.069	0.71
2 Equal	Excl. owner-operators (draft)	EEH hourly (cube 13)	339	0.127	0.058	0.100	0.72
2 Equal	Excl. owner-operators (draft)	ATO salary	342	0.122	0.039	0.086	0.71

Scheme	Sample	Ranking	Units	Claude	Copilot	Composite	Rank correlation
2 Equal	Excl. mostly unscored (added after results)	EEH weekly	332	0.097	0.025	0.065	0.72
2 Equal	Excl. mostly unscored (added after results)	EEH hourly (cube 13)	329	0.111	0.054	0.088	0.72
2 Equal	Excl. mostly unscored (added after results)	ATO salary	332	0.118	0.047	0.088	0.72
3 Title	All units	EEH weekly	352	0.065	−0.011	0.032	0.67
3 Title	All units	EEH hourly (cube 13)	349	0.074	0.025	0.055	0.67
3 Title	All units	ATO salary	352	0.087	0.012	0.055	0.67
3 Title	Non-diffuse	EEH weekly	214	0.099	−0.005	0.052	0.61
3 Title	Non-diffuse	EEH hourly (cube 13)	213	0.090	0.015	0.057	0.62
3 Title	Non-diffuse	ATO salary	214	0.130	0.029	0.089	0.61
3 Title	Excl. RSE above 25%	EEH weekly	327	0.067	−0.011	0.033	0.66
3 Title	Excl. RSE above 25%	EEH hourly (cube 13)	324	0.079	0.028	0.060	0.66
3 Title	Excl. RSE above 25%	ATO salary	327	0.086	0.010	0.054	0.66
3 Title	Excl. owner-operators (draft)	EEH weekly	339	0.071	−0.012	0.034	0.65
3 Title	Excl. owner-operators (draft)	EEH hourly (cube 13)	336	0.084	0.030	0.064	0.65
3 Title	Excl. owner-operators (draft)	ATO salary	339	0.089	0.005	0.053	0.65
3 Title	Excl. mostly unscored (added after results)	EEH weekly	329	0.065	0.002	0.040	0.69
3 Title	Excl. mostly unscored (added after results)	EEH hourly (cube 13)	326	0.075	0.036	0.062	0.70
3 Title	Excl. mostly unscored (added after results)	ATO salary	329	0.089	0.027	0.066	0.69
4 JSA links	All units	EEH weekly	347	0.043	−0.012	0.016	0.72
4 JSA links	All units	EEH hourly (cube 13)	345	0.059	0.018	0.040	0.73
4 JSA links	All units	ATO salary	347	0.061	0.012	0.039	0.72
4 JSA links	Non-diffuse	EEH weekly	208	0.098	0.019	0.064	0.68
4 JSA links	Non-diffuse	EEH hourly (cube 13)	208	0.099	0.033	0.073	0.68
4 JSA links	Non-diffuse	ATO salary	208	0.127	0.055	0.100	0.68
4 JSA links	Excl. RSE above 25%	EEH weekly	324	0.044	−0.014	0.015	0.72
4 JSA links	Excl. RSE above 25%	EEH hourly (cube 13)	322	0.062	0.020	0.043	0.72
4 JSA links	Excl. RSE above 25%	ATO salary	324	0.060	0.010	0.037	0.72
4 JSA links	Excl. owner-operators (draft)	EEH weekly	334	0.048	−0.014	0.016	0.72
4 JSA links	Excl. owner-operators (draft)	EEH hourly (cube 13)	332	0.070	0.018	0.046	0.72
4 JSA links	Excl. owner-operators (draft)	ATO salary	334	0.064	0.004	0.035	0.72
4 JSA links	Excl. mostly unscored (added after results)	EEH weekly	327	0.048	−0.011	0.018	0.72

Scheme	Sample	Ranking	Units	Claude	Copilot	Composite	Rank correlation
4 JSA links	Excl. mostly unscored (added after results)	EEH hourly (cube 13)	325	0.061	0.018	0.041	0.73
4 JSA links	Excl. mostly unscored (added after results)	ATO salary	327	0.067	0.014	0.043	0.72

14 Uncertainty: design and ranges

Added for v1.0, after the first results.

- **Crosswalk weights.** Each unit's scheme-1 weights p over its n scored US occupations are redrawn from Dirichlet$(\alpha n p)$. The concentration α is chosen from a grid so that the employment-weighted average spread of unit values across draws is closest to their spread across schemes 1 to 3. The wider setting uses $\alpha/4$.
- **Earnings.** Each unit's EEH weekly earnings are multiplied by $1 + \varepsilon$, with ε normal with standard deviation equal to the unit's published relative standard error (the median where none is published), independently across units. Ranks and deciles are re-formed in every draw.
- **Employment** is held fixed. EQ08 publishes no standard errors.
- **Draws.** 500 per setting, with a fixed seed, so reruns give identical results.

alpha	Spread across draws	Spread across schemes 1-3	Chosen
0.25	0.0269	0.0179	
0.50	0.0225	0.0179	
1.00	0.0162	0.0179	yes
2.00	0.0124	0.0179	
4.00	0.0089	0.0179	
8.00	0.0064	0.0179	
16.00	0.0046	0.0179	

Ranges over draws (5th, 50th and 95th percentiles).

Draws	Top minus bottom	CI	Rank CI
Crosswalk	1.5 / 4.2 / 7.1	0.065 / 0.089 / 0.113	0.060 / 0.070 / 0.082
Crosswalk, wider	−0.7 / 4.8 / 9.4	0.046 / 0.093 / 0.135	0.051 / 0.069 / 0.095
Earnings	2.6 / 3.7 / 5.1	0.078 / 0.089 / 0.101	0.067 / 0.073 / 0.080
Both	0.3 / 3.8 / 7.0	0.062 / 0.088 / 0.116	0.058 / 0.070 / 0.084

Bounds for unscored links. Lower: unscored US occupations count as zero. Upper: they count as one, the highest possible score. The high imputation gives them the highest score in their SOC minor group (major group if the minor group has none); it is not a bound, because for some units that score is below the mean of their scored links.

Version	Top minus bottom	CI	Rank CI	Peak decile	Holds under the rule
Lower bound (zero)	3.9	0.078	0.076	7	yes
Main (renormalised)	4.3	0.089	0.073	7	yes
High imputation	8.4	0.127	0.087	7	yes
Upper bound (one)	10.0	0.010	0.041	3	yes

Largest disagreements. Share of draws in which each of the fifteen largest sign-stable disagreements with JSA automation stays among the fifteen largest.

Unit	Title	Crosswalk	Both
5611	Betting Clerks	100%	100%
5912	Transport and Despatch Clerks	92%	92%
5911	Purchasing and Supply Logistics Clerks	98%	96%
1411	Cafe and Restaurant Managers	100%	100%
8991	Caretakers	100%	100%
5521	Bank Workers	100%	100%
1211	Aquaculture Farmers	100%	100%
7321	Delivery Drivers	90%	88%
2113	Photographers	97%	98%
7312	Bus and Coach Drivers	100%	100%
3422	Electrical Distribution Trades Workers	77%	77%
1334	Manufacturers	48%	53%
7313	Train and Tram Drivers	87%	86%
5513	Payroll Clerks	87%	85%
2114	Visual Arts and Crafts Professionals	40%	36%

15 Earnings, capability and computer work: full results

Added for v1.0, after the first results. Employment-weighted least squares on employment-weighted percentile ranks (0 to 100), HC1 standard errors, 352 units with every measure. Teleworkability is the classification by Dingel and Neiman (2020) of O*NET-SOC occupations, averaged to SOC 2010 codes and mapped to units through the SOC 2010 links and US employment weights used for AIOE. Its licence is a software licence (GPL-3.0), so no unit-level values are published.

Outcome	Controls	Term	Estimate	SE
Usage-based exposure	None	Earnings	0.222	0.091
Usage-based exposure	Teleworkability	Earnings	0.006	0.085
Usage-based exposure	Teleworkability	Teleworkability	0.630	0.070
Usage-based exposure	JSA automation	Earnings	0.151	0.053
Usage-based exposure	JSA automation	JSA automation	0.704	0.054
Usage-based exposure	JSA automation, teleworkability	Earnings	0.075	0.056
Usage-based exposure	JSA automation, teleworkability	JSA automation	0.554	0.077
Usage-based exposure	JSA automation, teleworkability	Teleworkability	0.267	0.080
Usage-based exposure	JSA augmentation	Earnings	−0.073	0.083
Usage-based exposure	JSA augmentation	JSA augmentation	0.667	0.068
Usage-based exposure	JSA augmentation, teleworkability	Earnings	−0.112	0.079
Usage-based exposure	JSA augmentation, teleworkability	JSA augmentation	0.448	0.084
Usage-based exposure	JSA augmentation, teleworkability	Teleworkability	0.395	0.090
Usage-based exposure	Eloundou beta	Earnings	−0.111	0.075
Usage-based exposure	Eloundou beta	Eloundou beta	0.814	0.042
Usage-based exposure	Eloundou beta, teleworkability	Earnings	−0.126	0.073
Usage-based exposure	Eloundou beta, teleworkability	Eloundou beta	0.689	0.075

Outcome	Controls	Term	Estimate	SE
Usage-based exposure	Eloundou beta, teleworkability	Teleworkability	0.193	0.082
Usage-based exposure	AIOE	Earnings	−0.142	0.066
Usage-based exposure	AIOE	AIOE	0.853	0.046
Usage-based exposure	AIOE, teleworkability	Earnings	−0.142	0.067
Usage-based exposure	AIOE, teleworkability	AIOE	0.844	0.069
Usage-based exposure	AIOE, teleworkability	Teleworkability	0.011	0.082
Extensive margin	None	Earnings	0.256	0.081
Extensive margin	Teleworkability	Earnings	0.064	0.076
Extensive margin	Teleworkability	Teleworkability	0.557	0.074
Extensive margin	JSA automation	Earnings	0.206	0.078
Extensive margin	JSA automation	JSA automation	0.491	0.078
Extensive margin	JSA automation, teleworkability	Earnings	0.099	0.084
Extensive margin	JSA automation, teleworkability	JSA automation	0.281	0.113
Extensive margin	JSA automation, teleworkability	Teleworkability	0.374	0.114
Extensive margin	JSA augmentation	Earnings	0.006	0.076
Extensive margin	JSA augmentation	JSA augmentation	0.563	0.084
Extensive margin	JSA augmentation, teleworkability	Earnings	−0.030	0.067
Extensive margin	JSA augmentation, teleworkability	JSA augmentation	0.358	0.109
Extensive margin	JSA augmentation, teleworkability	Teleworkability	0.370	0.100
Extensive margin	Eloundou beta	Earnings	−0.012	0.062
Extensive margin	Eloundou beta	Eloundou beta	0.653	0.060
Extensive margin	Eloundou beta, teleworkability	Earnings	−0.031	0.062
Extensive margin	Eloundou beta, teleworkability	Eloundou beta	0.496	0.103
Extensive margin	Eloundou beta, teleworkability	Teleworkability	0.243	0.106
Extensive margin	AIOE	Earnings	−0.053	0.065
Extensive margin	AIOE	AIOE	0.723	0.051
Extensive margin	AIOE, teleworkability	Earnings	−0.054	0.065
Extensive margin	AIOE, teleworkability	AIOE	0.676	0.087
Extensive margin	AIOE, teleworkability	Teleworkability	0.062	0.096

Gelbach decomposition. The change in the earnings slope when a potential measure and teleworkability are added, split between them (Gelbach 2016). The two parts add to the change.

Outcome	Potential measure	Slope without controls	Slope with controls	Through the potential measure	Through teleworkability
Usage-based exposure	JSA automation	0.222	0.075	0.056	0.091
Usage-based exposure	JSA augmentation	0.222	−0.112	0.198	0.135
Usage-based exposure	Eloundou beta	0.222	−0.126	0.282	0.066
Usage-based exposure	AIOE	0.222	−0.142	0.360	0.004

Outcome	Potential measure	Slope without controls	Slope with controls	Through the potential measure	Through teleworkability
Extensive margin	JSA automation	0.256	0.099	0.028	0.128
Extensive margin	JSA augmentation	0.256	−0.030	0.158	0.127
Extensive margin	Eloundou beta	0.256	−0.031	0.203	0.083
Extensive margin	AIOE	0.256	−0.054	0.288	0.021

Two measures of capability. JSA augmentation and AIOE treated as two error-prone measures of one capability; the two regressions are stacked and each measure is instrumented with the other (Gillen, Snowberg and Yariv 2019). Both regressions hold teleworkability fixed.

Outcome	Earnings	Capability	Teleworkability
Usage-based exposure	−0.268	1.040	0.085
Extensive margin	−0.154	0.832	0.122

16 Review of the forty largest units

Automated checks on the forty largest units (decision D10 in the decision log). The full table, with a resolution and reason for each flag, is `data/out/m2/top40_units_review.csv`.

Check	Units flagged
unmatched	1
zero	4
diffuse	10
scheme_range	9
influence	6
artefact	3
unscored	2
title	11

Unit	Title	Usage, scheme 1	Flags	Note
6211	Sales Assistants (General)	0.132	diffuse; scheme_range; influence; artefact; title	Retail Salespersons (a flagged artefact, 0.32) and Fast Food and Counter Workers (0.00) carry four-fifths of the weight; partial ISCO links in the ABS correspondence add small weights on beauty, teller and postal occupations. Largest single influence on the main profile; leave-one-out result reported.
2544	Registered Nurses	0.062	influence	
4231	Aged and Disabled Carers	0.042	unscored; title	The main US counterparts (Home Health Aides, Personal Care Aides) have no score, so the value rests on two supervisor occupations. Excluded in the unscored-majority sensitivity run.
5311	General Clerks	0.446	scheme_range; influence	

Unit	Title	Usage, scheme 1	Flags	Note
1421	Retail Managers	0.127	scheme_range	
2211	Accountants	0.329	scheme_range	
2613	Software and Applications Programmers	0.331	diffuse; scheme_range; influence	
3411	Electricians	0.037	diffuse	
7331	Truck Drivers	0.000	zero	
4211	Child Carers	0.021	title	Title flag is wording only (child carer / childcare worker); the link is the natural US counterpart.
5421	Receptionists	0.350	diffuse	
7411	Storepersons	0.002	diffuse; title	Title flag is wording only; US stockers and hand material movers are the natural counterparts.
1311	Advertising, Public Relations and Sales Managers	0.155	scheme_range; influence	
5111	Contract, Program and Project Administrators	0.177	diffuse; scheme_range; title	Closest US occupation (Project Management Specialists) has no score; value rests on business and financial specialist occupations (38% of linked US employment unscored).
8513	Kitchenhands	0.000	zero; title	Title flag is wording only; US food preparation workers are the natural counterpart.
5511	Accounting Clerks	0.284	influence	
4315	Waiters	0.000	zero	
4221	Education Aides	no score	unmatched; unscored	
3513	Chefs	0.048	title	Both links come from ISCO 3434; OEWS weights favour the larger US supervisor occupation over Chefs and Head Cooks.
3232	Metal Fitters and Machinists	0.027	diffuse	
4311	Bar Attendants and Baristas	0.000	zero; artefact; title	Single link to US Bartenders, a flagged artefact with a score of zero.
3212	Motor Mechanics	0.032	diffuse	
4233	Nursing Support and Personal Care Workers	0.009	diffuse	
2713	Solicitors	0.167	artefact; title	Title flag is wording only (solicitor / lawyer); US Lawyers is a flagged artefact.
2251	Advertising and Marketing Professionals	0.628	scheme_range	
1323	Human Resource Managers	0.062	scheme_range	
2531	General Practitioners and Resident Medical Officers	0.037	diffuse; title	Title flag is wording only; US physician occupations,

Unit	Title	Usage, scheme 1	Flags	Note
				including family medicine, are the counterparts.
5512	Bookkeepers	0.284	title	Title flag is wording only (bookkeeper / bookkeeping clerk).

17 Global automation share by decile

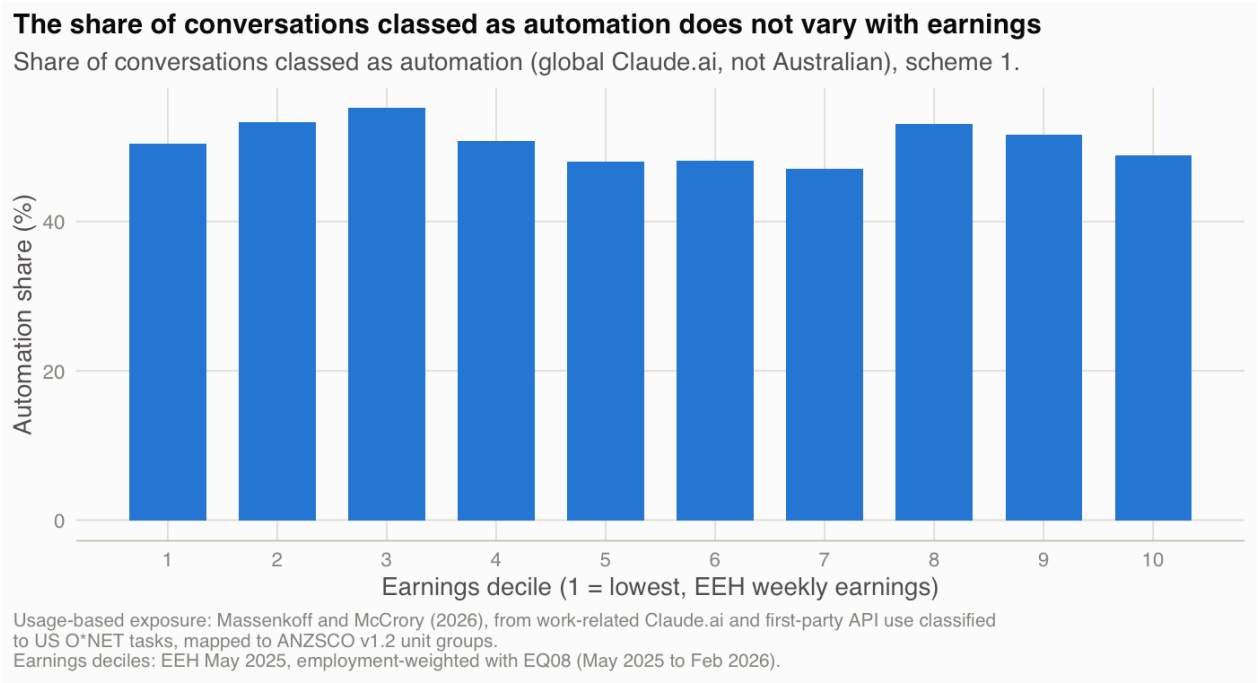


Figure 5: Share of global Claude.ai conversations classed as automation, averaged by earnings decile (scheme 1). These shares are global, not Australian.

18 Rebuilding the usage-based score from task data

The published occupation scores were rebuilt from the published task-level values, using O*NET task weights in place of the time fractions Massenkoff and McCrory use, which are not published. The rank correlation with the published scores falls short of the 0.9 target. The correlation among occupations with a positive published score (added after the first results) shows how much of the agreement comes from matching zeros. The main analysis uses the published scores.

Task weights	SOC codes compared	Spearman	Agreement on exact zeros	Positive scores	Spearman, positive scores	Meets target
equal	756	0.871	93%	345	0.816	no
importance	748	0.869	93%	342	0.827	no
importance × relevance	748	0.870	93%	342	0.840	no

19 The two-capital model: premium and benchmark responses

The main text states the responses of the skill premium to ICT, machinery and AI capital. With competitive wages, the skill premium in the thesis's model is

$$\pi = c_\pi \left(\frac{V}{X}\right)^{1/\sigma_{VX}} \left(\frac{X}{S}\right)^{1/\sigma_{XS}} \left(\frac{U}{V}\right)^{1/\sigma_{VU}} \frac{A^S}{A^U},$$

with $s_I = (1-\mu)(K^I)^{\eta_X}/X^{\eta_X}$ ICT's cost share in the skilled branch, $s_S = 1 - s_I$, and s_M and $s_U = 1 - s_M$ defined likewise in the unskilled branch. Differentiating $\ln \pi$ with respect to $\ln K^I$ and $\ln K^M$ gives the benchmark responses in the main text; substituting $K^I + \zeta K^A$ for K^I (scenario C) or $A^S N^S + \psi K^A$ for S (scenario D) and applying the chain rule gives the two AI responses. If AI instead performs unskilled tasks, $U \to A^U N^U + \psi K^A$, then

$$\frac{\partial \ln \pi}{\partial \ln K^A} = \frac{\psi K^A}{U}\left[\frac{s_U}{\sigma_{VX}} + \frac{1 - s_U}{\sigma_{VU}}\right] > 0.$$

Every expression is checked symbolically and numerically in `theory/check_theory.py`, with output in `theory/check_theory_output.txt`. The responses hold technology fixed; how directed technical change would respond to AI is not derived.

20 Data sources and licences

Commit hashes are shortened to twelve characters; `data/raw/MANIFEST.csv` gives them in full, with each file's URL and checksum.

Source	Version	Licence
aei_card	4e5f69fab13a	CC-BY (version not stated on dataset card); HF metadata tag: MIT
aei_job_exposure	4e5f69fab13a	CC-BY (version not stated on dataset card); HF metadata tag: MIT
aei_task_penetration	4e5f69fab13a	CC-BY (version not stated on dataset card); HF metadata tag: MIT
aei_claude_ai_2026_06_26	4e5f69fab13a	CC-BY (version not stated on dataset card); HF metadata tag: MIT
aei_doc_2026_06_26	4e5f69fab13a	CC-BY (version not stated on dataset card); HF metadata tag: MIT
mm_appendix	5 Mar 2026	Not stated
onet_task_statements	O*NET 31.0	CC BY 4.0
onet_task_ratings	O*NET 31.0	CC BY 4.0
onet_occupation_data	O*NET 31.0	CC BY 4.0
abs_v12_isco08	ANZSCO 2013 v1.2, released 4 Oct 2013	CC BY 4.0
abs_v12_structure	ANZSCO 2013 v1.2	CC BY 4.0
abs_v13_isco08	ANZSCO 2013 v1.3, released 5 Nov 2019	CC BY 4.0
abs_v13_structure	ANZSCO 2013 v1.3	CC BY 4.0
abs_v13_whatsnew	ANZSCO 2013 v1.3	CC BY 4.0
abs_2021_correspondence	ANZSCO 2021	CC BY 4.0
abs_2021_structure	ANZSCO 2021	CC BY 4.0
abs_2022_correspondence	ANZSCO 2022	CC BY 4.0
abs_2022_structure	ANZSCO 2022	CC BY 4.0
abs_eq08	Labour Force, Australia, Detailed, March 2026	CC BY 4.0
abs_eeh_cubes	Employee Earnings and Hours, May 2025 (released 23 Jan 2026)	CC BY 4.0
abs_wa_table04	Working arrangements, August 2025, Table 4	CC BY 4.0
ato_ts24_table15	Taxation Statistics 2023-24, Individuals Table 15	CC BY 2.5 AU
jsa_occupations_8	Our Gen AI Transition (14 Aug 2025)	CC BY 4.0
eloundou_occ_level	0471612fef3c	MIT

Source	Version	Licence
aioe_appendix	adca5fc2cd0e	None stated
aioe_lm	adca5fc2cd0e	None stated
lmi_anzsco_onetsoc	Internet Archive capture 15 Mar 2024; ANZSCO 2013 v1.2 (per the file's Notes sheet)	CC BY 4.0
wwai_applicability	550c00c9d1b0	CC BY 4.0
wwai_soc_metrics	550c00c9d1b0	CC BY 4.0
wwai_licence	550c00c9d1b0	CC BY 4.0
dn_workathome	3ebab4f646ed	GPL-3.0
dn_licence	3ebab4f646ed	GPL-3.0
bls_soc_2010_2018	SOC 2010 to 2018	US Government work, public domain
bls_isco_soc	ISCO-08 to SOC 2010	US Government work, public domain
bls_oews_2025	OEWS May 2025 national	US Government work, public domain

21 Edition bridge: ANZSCO 2022 to v1.2

Unit-group pairs that do not keep the same code. The full table is `data/interim/bridge_2022units_v12units.csv`.

2022 unit	v1.2 unit	v1.2 title	Six-digit links	Type
1215	1212	Crop Farmers	4	merge (several 2022 units into one v1.2 unit)
1216	1212	Crop Farmers	10	merge (several 2022 units into one v1.2 unit)
1217	1212	Crop Farmers	1	many-to-many
1217	1214	Mixed Crop and Livestock Farmers	2	split (one 2022 unit into several v1.2 units)
2233	2232	ICT Trainers	1	split (one 2022 unit into several v1.2 units)
2233	2233	Training and Development Professionals	1	split (one 2022 unit into several v1.2 units)
2254	2254	Technical Sales Representatives	3	merge (several 2022 units into one v1.2 unit)
2341	2254	Technical Sales Representatives	1	many-to-many
2341	2341	Agricultural and Forestry Scientists	5	split (one 2022 unit into several v1.2 units)
2341	2345	Life Scientists	2	many-to-many
2345	2345	Life Scientists	9	merge (several 2022 units into one v1.2 unit)
2346	2346	Medical Laboratory Scientists	1	split (one 2022 unit into several v1.2 units)
2346	2349	Other Natural and Physical Science Professionals	1	many-to-many
2349	2349	Other Natural and Physical Science Professionals	6	merge (several 2022 units into one v1.2 unit)
2611	2611	ICT Business and Systems Analysts	2	split (one 2022 unit into several v1.2 units)
2611	2632	ICT Support and Test Engineers	1	many-to-many
2632	2632	ICT Support and Test Engineers	4	merge (several 2022 units into one v1.2 unit)
3625	3622	Gardeners	2	merge (several 2022 units into one v1.2 unit)

2022 unit	v1.2 unit	v1.2 title	Six-digit links	Type
3626	3622	Gardeners	1	merge (several 2022 units into one v1.2 unit)
3627	3622	Gardeners	2	merge (several 2022 units into one v1.2 unit)
3633	3612	Shearers	1	split (one 2022 unit into several v1.2 units)
3633	3999	Other Miscellaneous Technicians and Trades Workers	1	many-to-many
3999	3999	Other Miscellaneous Technicians and Trades Workers	8	merge (several 2022 units into one v1.2 unit)
8421	8411	Aquaculture Workers	1	recoded one-to-one
8422	8412	Crop Farm Workers	12	recoded one-to-one
8423	8415	Livestock Farm Workers	10	recoded one-to-one
8424	8416	Mixed Crop and Livestock Farm Workers	2	recoded one-to-one
8431	8413	Forestry and Logging Workers	3	recoded one-to-one
8432	8414	Garden and Nursery Labourers	1	merge (several 2022 units into one v1.2 unit)
8433	8414	Garden and Nursery Labourers	1	merge (several 2022 units into one v1.2 unit)
8434	8419	Other Farm, Forestry and Garden Workers	1	merge (several 2022 units into one v1.2 unit)
8439	8419	Other Farm, Forestry and Garden Workers	3	merge (several 2022 units into one v1.2 unit)

22 Earnings imputations

Units without a published EEH value after bridging, and the parent level used. “none” means the unit has a published value in that source; “not imputed” means it has none and none was imputed (EEH cube 13 covers non-managerial employees only, so the chief executive, general manager and legislator units drop out of the hourly ranking).

Unit	Title	EEH weekly	EEH cube 13 hourly	ATO
1111	Chief Executives and Managing Directors	none	not imputed	none
1112	General Managers	none	not imputed	none
1113	Legislators	minor group (3-digit) mean	not imputed	none
1211	Aquaculture Farmers	minor group (3-digit) mean	none	none
1212	Crop Farmers	minor group (3-digit) mean	none	none
1214	Mixed Crop and Livestock Farmers	minor group (3-digit) mean	none	none
1392	Senior Non-commissioned Defence Force Members	minor group (3-digit) mean	none	none
2522	Complementary Health Therapists	minor group (3-digit) mean	none	none
3221	Metal Casting, Forging and Finishing Trades Workers	minor group (3-digit) mean	none	none
3612	Shearers	minor group (3-digit) mean	none	none
3931	Canvas and Leather Goods Makers	minor group (3-digit) mean	none	none
3994	Jewellers	minor group (3-digit) mean	none	none
3996	Signwriters	minor group (3-digit) mean	none	none

Unit	Title	EEH weekly	EEH cube 13 hourly	ATO
4411	Defence Force Members - Other Ranks	minor group (3-digit) mean	none	none
4513	Funeral Workers	minor group (3-digit) mean	none	none
4515	Personal Care Consultants	minor group (3-digit) mean	none	none
5614	Mail Sorters	minor group (3-digit) mean	none	none
6111	Auctioneers, and Stock and Station Agents	minor group (3-digit) mean	none	none
6217	Street Vendors and Related Salespersons	minor group (3-digit) mean	none	none
6399	Other Sales Support Workers	none	none	minor group (3-digit) mean
8411	Aquaculture Workers	minor group (3-digit) mean	none	none
8413	Forestry and Logging Workers	minor group (3-digit) mean	none	none

23 Units without usage-based exposure

Unit	Title	Employment ('000)	Share of employment
4221	Education Aides	132.5	0.90%
6392	Retail and Wool Buyers	6.1	0.04%
1392	Senior Non-commissioned Defence Force Members	0.3	0.00%

24 Primary chain against the 2022-path chain

Units whose linked US occupations differ between the two chains.

Unit	Title	Primary links	2022-path links	Jaccard	Only in primary	Only in 2022 path
1212	Crop Farmers	5	6	0.83		45-2021
2254	Technical Sales Representatives	2	14	0.14		19-1011; 19-1012; 19-1013; 19-1021; 19-1022; 19-1023; 19-1029; 19-1041; 19-1042; 19-1099; 19-1032; 25-9021
2341	Agricultural and Forestry Scientists	3	14	0.21		19-1011; 19-1012; 19-1021; 19-1022; 19-1023; 19-1029; 19-1041; 19-1042; 19-1099; 41-1012; 41-4011
2345	Life Scientists	10	14	0.71		19-1032; 25-9021; 41-1012; 41-4011
3321	Floor Finishers	4	7	0.57		47-2051; 47-2053; 47-4091
4111	Ambulance Officers and Paramedics	10	11	0.91		29-1071
8415	Livestock Farm Workers	4	2	0.50	45-1011; 45-2021	
8419	Other Farm, Forestry and Garden Workers	5	6	0.83		53-7062

25 US employment weights not taken directly from OEWS

SOC	Title	Weight	Source
13-1021	Buyers and Purchasing Agents, Farm Products	163,810	OEWS broad group 13-1020 residual, split equally
13-1022	Wholesale and Retail Buyers, Except Farm Products	163,810	OEWS broad group 13-1020 residual, split equally

SOC	Title	Weight	Source
13-1023	Purchasing Agents, Except Wholesale, Retail, and Farm Products	163,810	OEWS broad group 13-1020 residual, split equally
13-2022	Appraisers of Personal and Business Property	28,535	OEWS broad group 13-2020 residual, split equally
13-2023	Appraisers and Assessors of Real Estate	28,535	OEWS broad group 13-2020 residual, split equally
21-1011	Substance Abuse and Behavioral Disorder Counselors	245,965	OEWS broad group 21-1010 residual, split equally
21-1014	Mental Health Counselors	245,965	OEWS broad group 21-1010 residual, split equally
25-2055	Special Education Teachers, Kindergarten	130,430	OEWS broad group 25-2050 residual, split equally
25-2056	Special Education Teachers, Elementary School	130,430	OEWS broad group 25-2050 residual, split equally
25-9042	Teaching Assistants, Preschool, Elementary, Middle, and Secondary School, Except Special Education	473,450	OEWS broad group 25-9040 residual, split equally
25-9043	Teaching Assistants, Special Education	473,450	OEWS broad group 25-9040 residual, split equally
25-9049	Teaching Assistants, All Other	473,450	OEWS broad group 25-9040 residual, split equally
29-2011	Medical and Clinical Laboratory Technologists	166,470	OEWS broad group 29-2010 residual, split equally
29-2012	Medical and Clinical Laboratory Technicians	166,470	OEWS broad group 29-2010 residual, split equally
31-1121	Home Health Aides	2,152,905	OEWS broad group 31-1120 residual, split equally
31-1122	Personal Care Aides	2,152,905	OEWS broad group 31-1120 residual, split equally
39-7011	Tour Guides and Escorts	26,750	OEWS broad group 39-7010 residual, split equally
39-7012	Travel Guides	26,750	OEWS broad group 39-7010 residual, split equally
47-4091	Segmental Pavers	14,190	OEWS broad group 47-4090 residual, split equally
47-4099	Construction and Related Workers, All Other	14,190	OEWS broad group 47-4090 residual, split equally
51-2022	Electrical and Electronic Equipment Assemblers	123,485	OEWS broad group 51-2020 residual, split equally
51-2023	Electromechanical Equipment Assemblers	123,485	OEWS broad group 51-2020 residual, split equally
51-2092	Team Assemblers	702,515	OEWS broad group 51-2090 residual, split equally
51-2099	Assemblers and Fabricators, All Other	702,515	OEWS broad group 51-2090 residual, split equally
53-1042	First-Line Supervisors of Helpers, Laborers, and Material Movers, Hand	155,910	OEWS broad group 53-1040 residual, split equally
53-1043	First-Line Supervisors of Material-Moving Machine and Vehicle Operators	155,910	OEWS broad group 53-1040 residual, split equally
53-1044	First-Line Supervisors of Passenger Attendants	155,910	OEWS broad group 53-1040 residual, split equally
53-1049	First-Line Supervisors of Transportation Workers, All Other	155,910	OEWS broad group 53-1040 residual, split equally
11-1031	Legislators	5,030	no OEWS row; smallest published detailed employment in its SOC major group
45-3031	Fishing and Hunting Workers	1,330	no OEWS row; smallest published detailed employment in its SOC major group

US military occupations (SOC major group 55) have no OEWS employment and no usage-based exposure, so they carry no weight.

26 AEI occupation-group shares: published mass and suppressed cells

Geography	Month	Published share (%)	Groups published
AU-WA	2026-04	97.46	18

Geography	Month	Published share (%)	Groups published
AU-WA	2026-05	98.67	19
AUS	2026-04	97.63	22
AUS	2026-05	98.90	22
GLOBAL	2026-04	94.58	22
GLOBAL	2026-05	98.74	22

Geography	Month	SOC major group	Name
AU-WA	2026-04	37	Building and Grounds Cleaning and Maintenance
AU-WA	2026-04	45	Farming, Fishing, and Forestry
AU-WA	2026-04	47	Construction and Extraction
AU-WA	2026-04	53	Transportation and Material Moving
AU-WA	2026-05	37	Building and Grounds Cleaning and Maintenance
AU-WA	2026-05	45	Farming, Fishing, and Forestry
AU-WA	2026-05	47	Construction and Extraction

27 Draft list of owner-operator units

Excluded in one sensitivity run. This is a judgement list awaiting review (see the Methods section).

Unit	Title
1211	Aquaculture Farmers
1212	Crop Farmers
1213	Livestock Farmers
1214	Mixed Crop and Livestock Farmers
1411	Cafe and Restaurant Managers
1413	Hotel and Motel Managers
1421	Retail Managers
3312	Carpenters and Joiners
3322	Painting Trades Workers
3341	Plumbers
3411	Electricians
3911	Hairdressers
7311	Automobile Drivers